

ACTIVITY

TOOL: R Programming

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Slot: B

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COURSE: Data Handling and Visualization

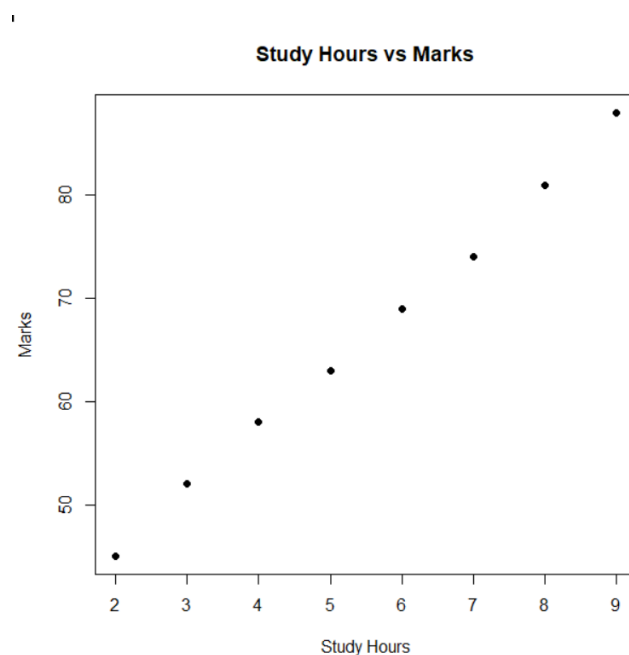
Code: DSA0604

Q1. Study Hours vs Marks – R

CODE:

```
data <- data.frame(  
  StudyHours = c(2,3,4,5,6,7,8,9),  
  Marks = c(45,52,58,63,69,74,81,88)  
)  
plot(data$StudyHours, data$Marks,  
  main = "Study Hours vs Marks",  
  xlab = "Study Hours",  
  ylab = "Marks",  
  pch = 19)  
cor(data$StudyHours, data$Marks)
```

OUTPUT:



Q2. Hospital – Age vs Systolic Blood Pressure

CODE:

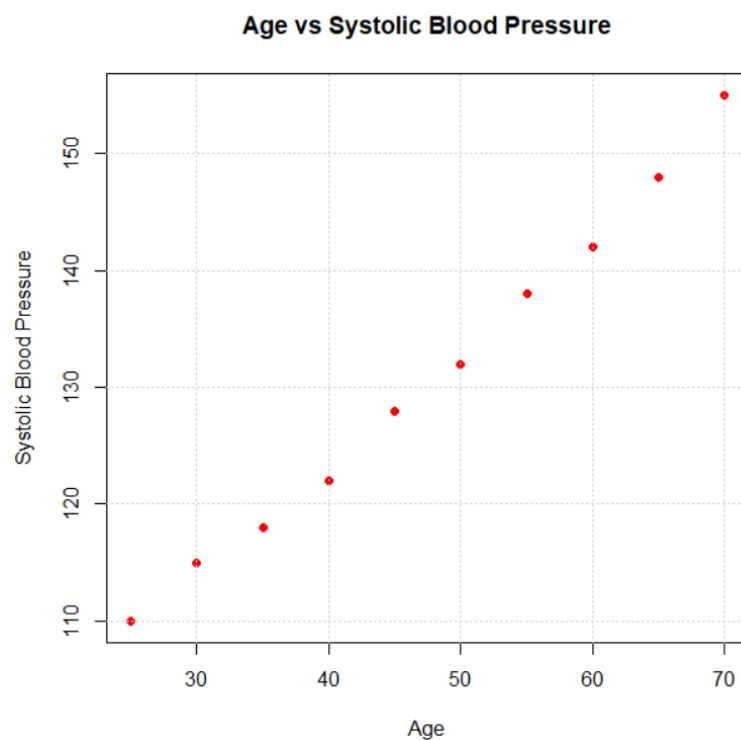
```
age <- c(25, 30, 35, 40, 45, 50, 55, 60, 65, 70)
```

```
blood_pressure <- c(110, 115, 118, 122, 128,  
                    132, 138, 142, 148, 155)
```

```
plot(age, blood_pressure,  
     main = "Age vs Systolic Blood Pressure",  
     xlab = "Age",  
     ylab = "Systolic Blood Pressure",  
     col = "red",  
     pch = 19)
```

```
grid()
```

OUTPUT:



Q3. Advertising Expenditure vs Monthly Sales

CODE:

```
advertising <- c(10, 15, 20, 25, 30, 35, 40, 45, 50, 55)
```

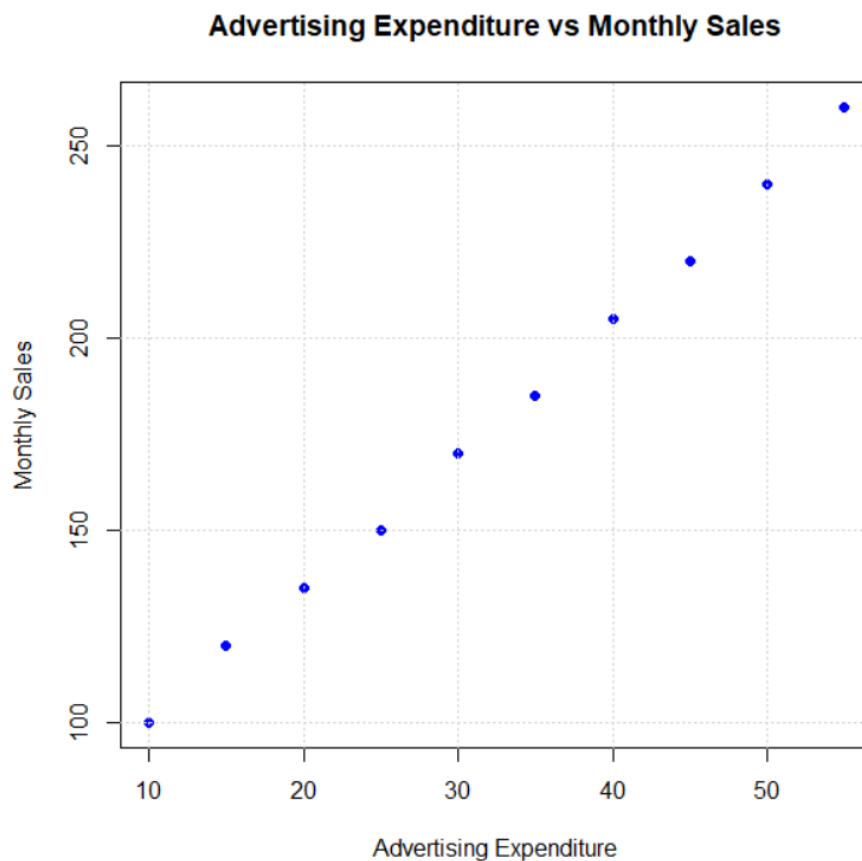
```
sales <- c(100, 120, 135, 150, 170, 185, 205, 220, 240, 260)
```

```

plot(advertising, sales,
     main = "Advertising Expenditure vs Monthly Sales",
     xlab = "Advertising Expenditure",
     ylab = "Monthly Sales",
     col = "blue",
     pch = 19)
grid()
correlation <- cor(advertising, sales)
print(paste("Correlation =", correlation))

```

OUTPUT:



Q4. Financial Institution – Correlation Matrix and Correlogram

CODE:

```

data <- data.frame(
  Income = c(30, 35, 40, 45, 50, 55, 60, 65, 70, 75),

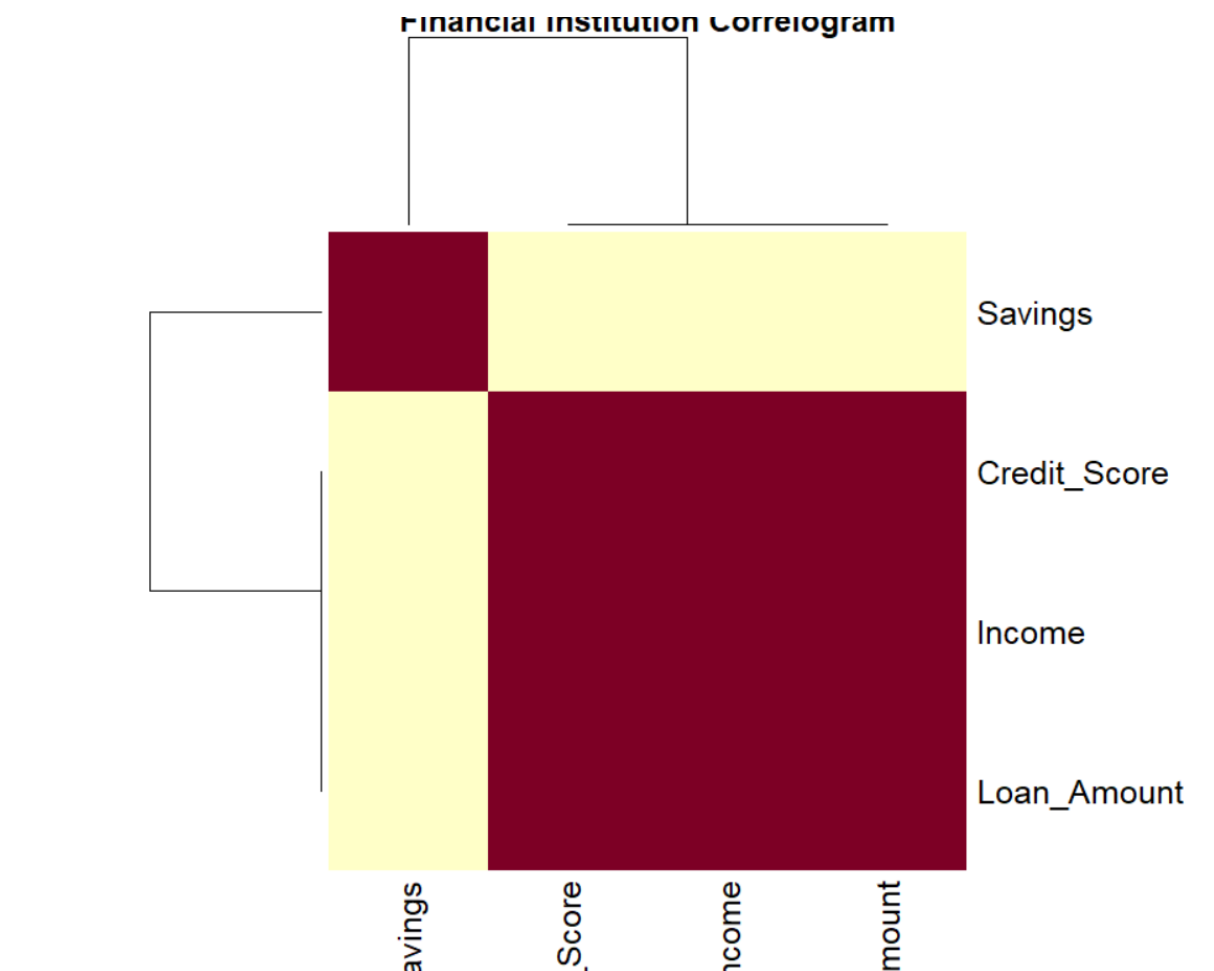
```

```

Savings = c(8, 10, 12, 14, 17, 19, 22, 24, 27, 30),
Loan_Amount = c(20, 25, 30, 35, 40, 45, 50, 55, 60, 65),
Credit_Score = c(650, 660, 670, 680, 690, 700, 710, 720, 730, 740)
)
cor_matrix <- cor(data)
print("Correlation Matrix:")
print(cor_matrix)
heatmap(cor_matrix,
        main = "Financial Institution Correlogram",
        symm = TRUE)

```

OUTPUT:



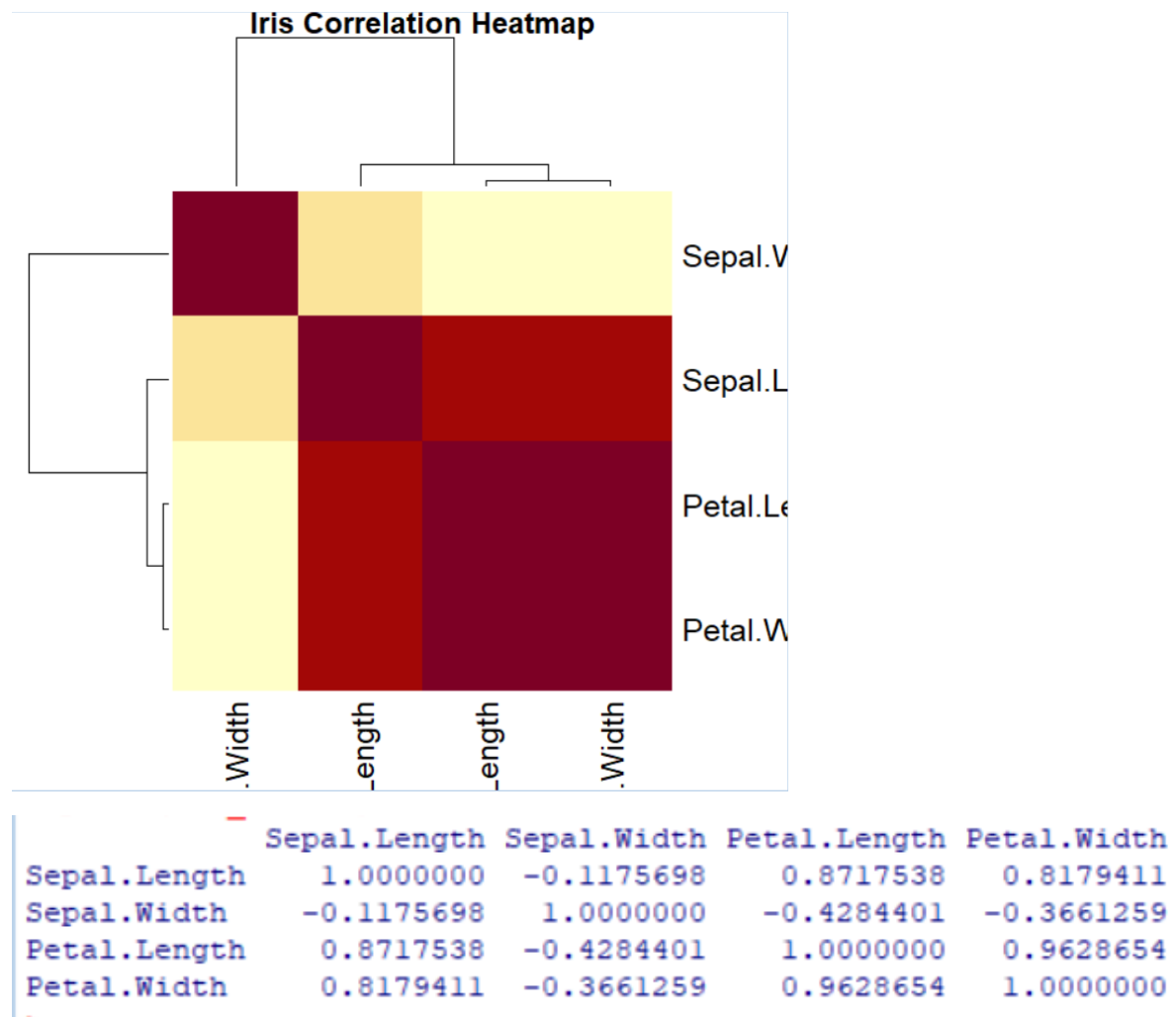
Q5. Iris Dataset – Correlation Matrix and Heatmap

CODE:

```
data(iris)
```

```
x <- iris[, 1:4]
cor_matrix <- cor(x)
print("Correlation Matrix:")
print(cor_matrix)
heatmap(cor_matrix,
        main = "Iris Correlation Heatmap",
        symm = TRUE)
```

OUTPUT:



Q6. Healthcare Dataset – Correlogram

CODE:

```
health <- data.frame(
  BMI = c(22,25,28,30,32,24,27,29,31,26),
  Cholesterol = c(180,190,210,230,240,185,205,220,235,200),
```

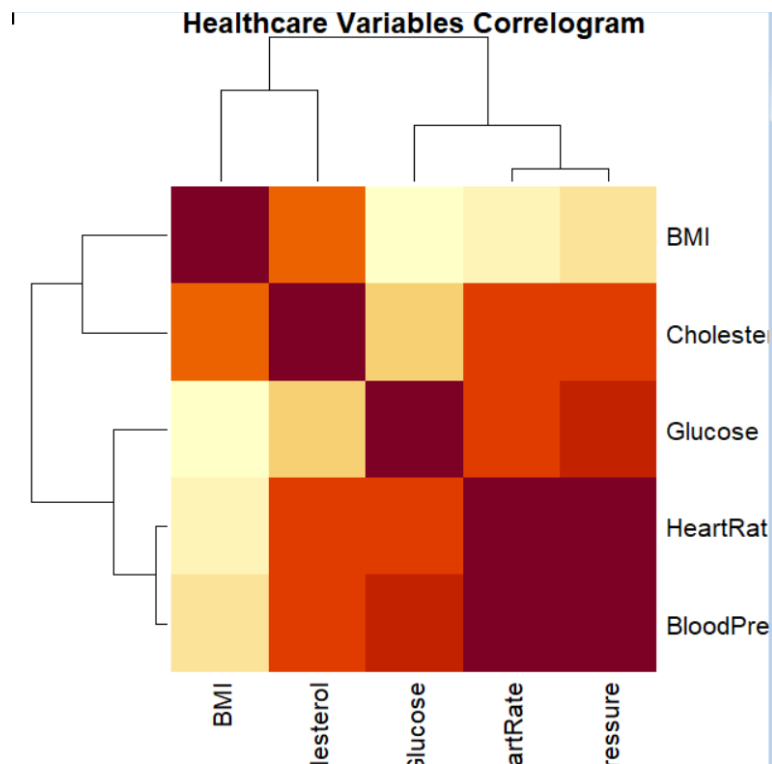
```

BloodPressure = c(110,115,120,130,135,112,118,125,132,117),
Glucose = c(90,100,110,125,140,95,105,115,130,102),
HeartRate = c(70,72,75,80,82,71,74,77,81,73)
)

cor_matrix <- cor(health)
print(cor_matrix)
heatmap(cor_matrix,
        main = "Healthcare Variables Correlogram",
        symm = TRUE)

```

OUTPUT:



Q7. MRI Features – PCA

CODE:

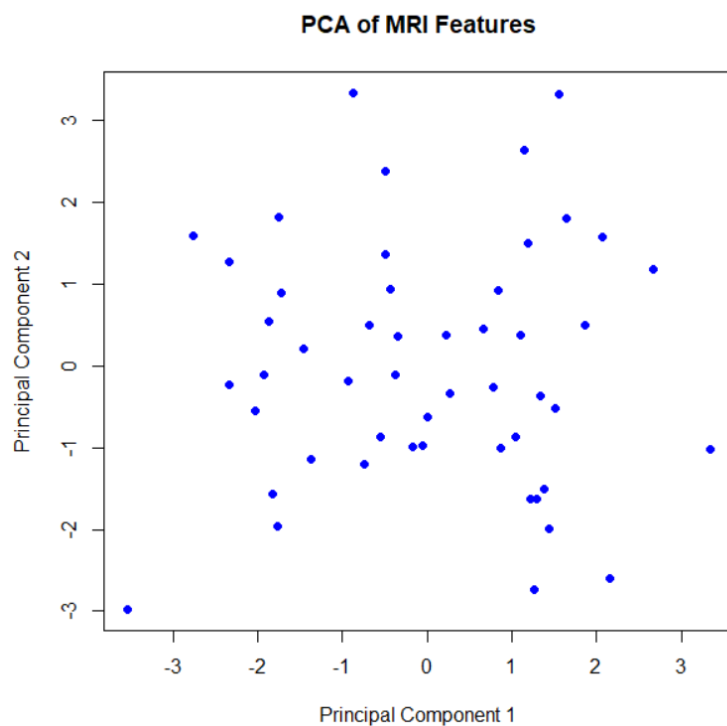
```

set.seed(123)
mri <- matrix(
  rnorm(25 * 50),
  nrow = 50,
  ncol = 25
)

```

```
)
pca <- prcomp(mri, scale. = TRUE)
pca_2d <- pca$x[, 1:2]
print(head(pca_2d))
plot(pca_2d[,1], pca_2d[,2],
     main = "PCA of MRI Features",
     xlab = "Principal Component 1",
     ylab = "Principal Component 2",
     col = "blue",
     pch = 19)
```

OUTPUT:



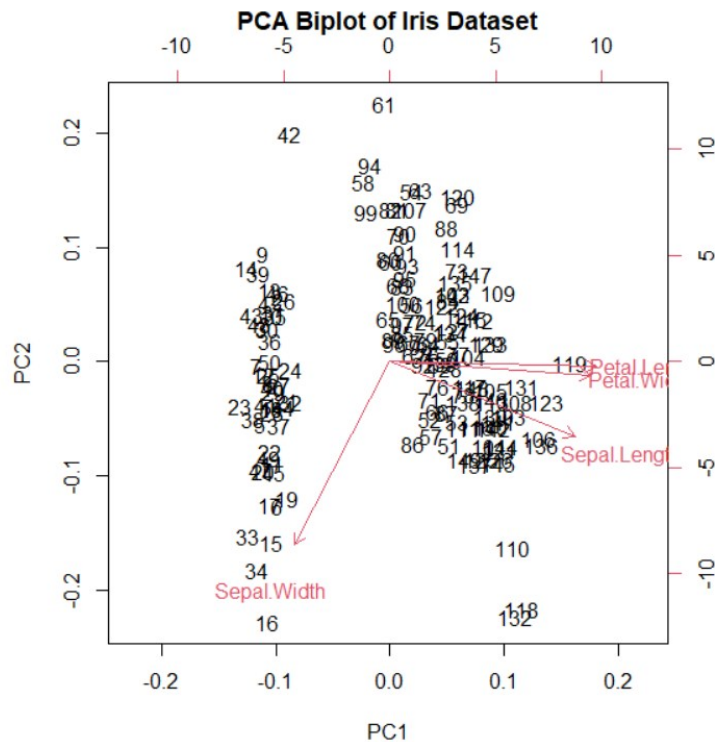
Q8. Iris Dataset – PCA and Biplot

CODE:

```
data(iris)
x <- iris[, 1:4]
pca <- prcomp(x, scale. = TRUE)
print(summary(pca))
```

```
biplot(pca,
       main = "PCA Biplot of Iris Dataset")
```

OUTPUT:



Answer / Interpretation

PC1 explains the highest variance among all the principal components. It captures the largest amount of information from the original four Iris variables.

The PCA transforms the original variables into new uncorrelated principal components.

Q9. Automobile Dataset – PCA

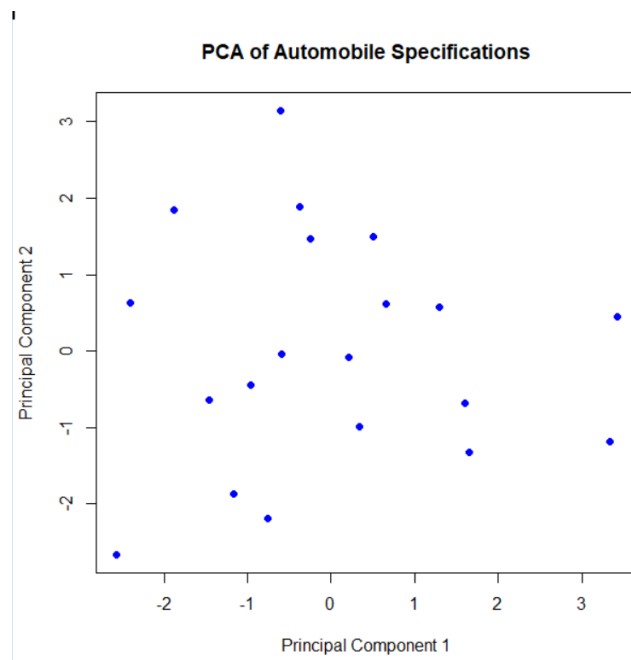
CODE:

```
set.seed(123)
cars <- matrix(
  rnorm(20 * 15),
  nrow = 20,
  ncol = 15
)
pca <- prcomp(cars, scale. = TRUE)
pc <- pca$x[, 1:2]
```



```
plot(pc[,1], pc[,2],
     main = "PCA of Automobile Specifications",
     xlab = "Principal Component 1",
     ylab = "Principal Component 2",
     col = "blue",
     pch = 19)
```

OUTPUT:



Q10. Facial Recognition – 128-Dimensional Embeddings using t-SNE

CODE:

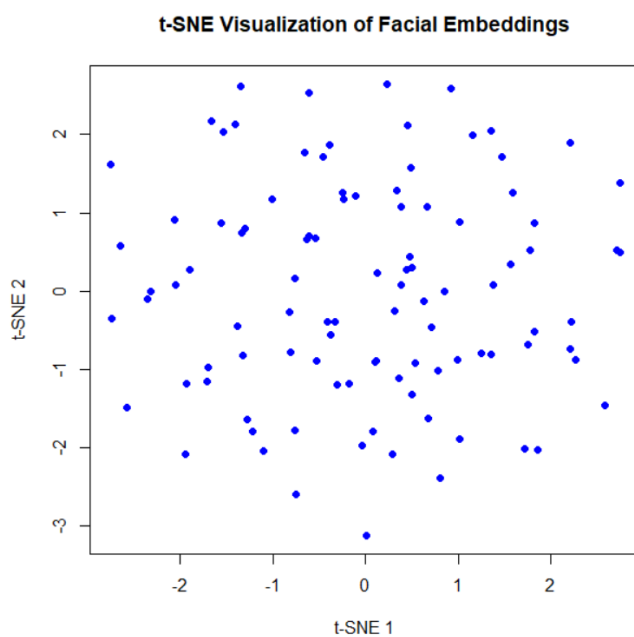
```
library(Rtsne)
set.seed(123)
data <- matrix(
  rnorm(100 * 128),
  nrow = 100,
  ncol = 128
)
tsne_result <- Rtsne(
  data,
```

```

dims = 2,
perplexity = 30,
verbose = FALSE
)
plot(tsne_result$Y,
     main = "t-SNE Visualization of Facial Embeddings",
     xlab = "t-SNE 1",
     ylab = "t-SNE 2",
     col = "blue",
     pch = 19)

```

OUTPUT:



Why t-SNE is preferred over PCA?

t-SNE is preferred for visualization because it is effective at preserving local relationships between high-dimensional data points. It can reveal clusters of similar facial embeddings more clearly in two dimensions, whereas PCA mainly preserves the directions of maximum overall variance.

Q11. Iris Dataset – t-SNE

CODE:

```

library(Rtsne)
data(iris)

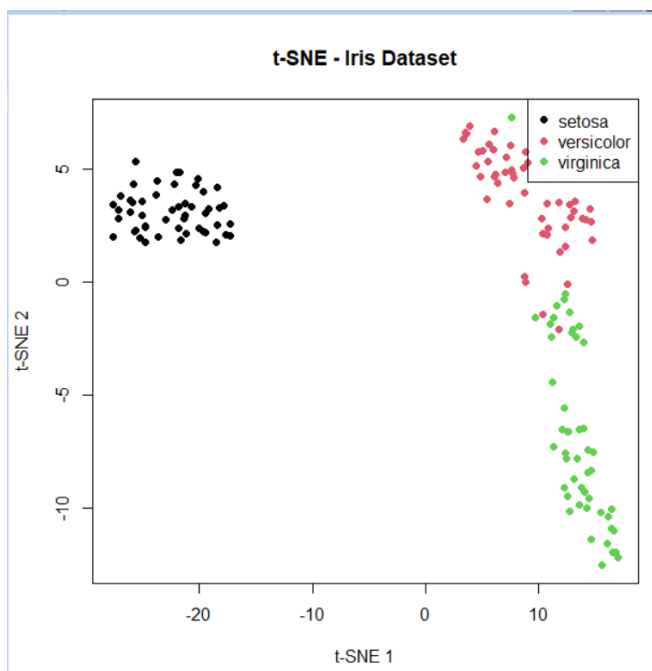
```

```

x <- iris[, 1:4]
set.seed(123)
tsne <- Rtsne(x,
  dims = 2,
  perplexity = 20,
  check_duplicates = FALSE)
plot(tsne$Y[,1], tsne$Y[,2],
  col = as.numeric(iris$Species),
  pch = 19,
  main = "t-SNE - Iris Dataset",
  xlab = "t-SNE 1",
  ylab = "t-SNE 2")
legend("topright",
  legend = levels(iris$Species),
  col = 1:3,
  pch = 19)

```

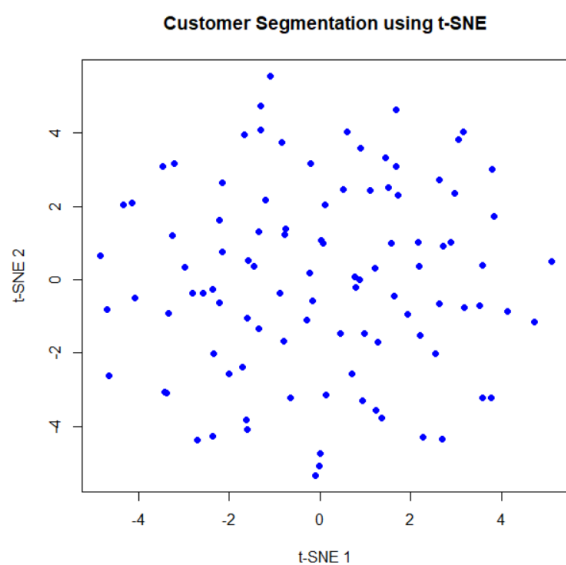
OUTPUT:



Q12. Customer Segmentation using t-SNE

CODE:

```
install.packages("Rtsne")  
library(Rtsne)  
set.seed(123)  
customers <- matrix(  
  rnorm(100 * 40),  
  nrow = 100,  
  ncol = 40  
)  
tsne <- Rtsne(customers,  
  dims = 2,  
  perplexity = 20,  
  check_duplicates = FALSE)  
plot(tsne$Y[,1], tsne$Y[,2],  
  col = "blue",  
  pch = 19,  
  main = "Customer Segmentation using t-SNE",  
  xlab = "t-SNE 1",  
  ylab = "t-SNE 2")
```

OUTPUT:

Answer / Discussion

t-SNE reduces the 40 behavioral variables into two dimensions. Points that are close together represent customers with similar behavior. Clearly separated groups indicate distinct customer clusters. If the points are highly mixed, distinct clusters are not clearly formed.

Q13. Medical Image Dataset – UMAP and PCA

CODE:

```
library(umap)

set.seed(123)

medical <- matrix(
  rnorm(100 * 100),
  nrow = 100,
  ncol = 100
)

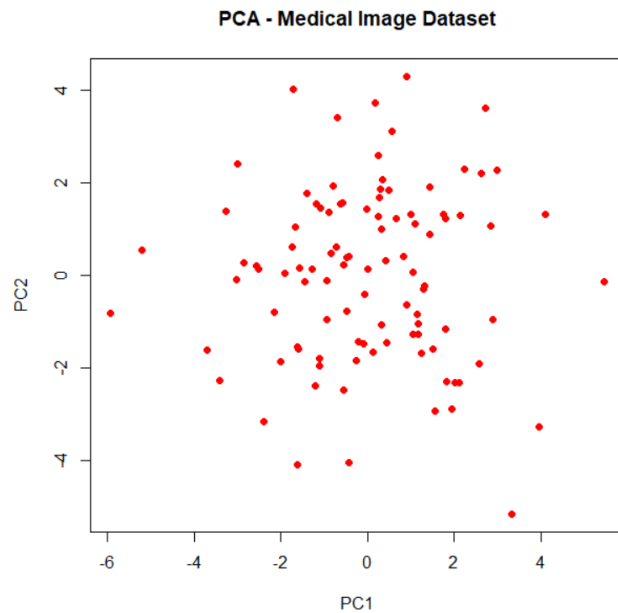
umap_result <- umap(medical)

plot(umap_result$layout[,1],
     umap_result$layout[,2],
     col = "blue",
     pch = 19,
     main = "UMAP - Medical Image Dataset",
     xlab = "UMAP 1",
     ylab = "UMAP 2")

pca_result <- prcomp(medical, scale. = TRUE)

plot(pca_result$x[,1],
     pca_result$x[,2],
     col = "red",
     pch = 19,
     main = "PCA - Medical Image Dataset",
     xlab = "PC1",
     ylab = "PC2")
```

OUTPUT:



Answer / Comparison

UMAP is useful for preserving local relationships and identifying clusters in high-dimensional data. PCA preserves the directions of maximum variance. UMAP can reveal nonlinear patterns that may not be visible using PCA.

Q14. Iris Dataset – UMAP

CODE:

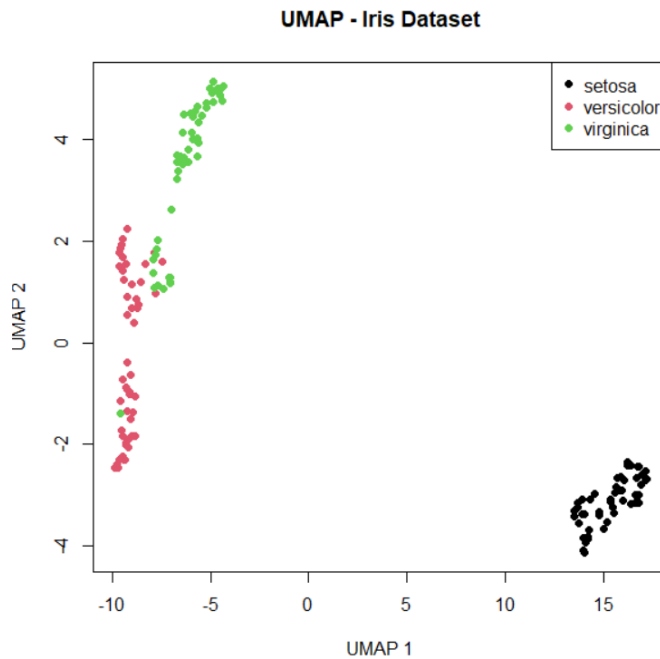
```
install.packages("umap")
library(umap)
data(iris)
x <- iris[,1:4]
set.seed(123)
result <- umap(x)
plot(result$layout[,1],
      result$layout[,2],
      col = as.numeric(iris$Species),
      pch = 19,
      main = "UMAP - Iris Dataset",
      xlab = "UMAP 1",
      ylab = "UMAP 2")
```

```

legend("topright",
      legend = levels(iris$Species),
      col = 1:3,
      pch = 19)

```

OUTPUT:



Answer / Interpretation

UMAP reduces the four Iris features into two dimensions. The three species generally form separate clusters. Setosa is usually clearly separated, while Versicolor and Virginica may have some overlap.

Q15. E-Commerce Customer Behavior – UMAP

CODE:

```

install.packages("umap")

library(umap)

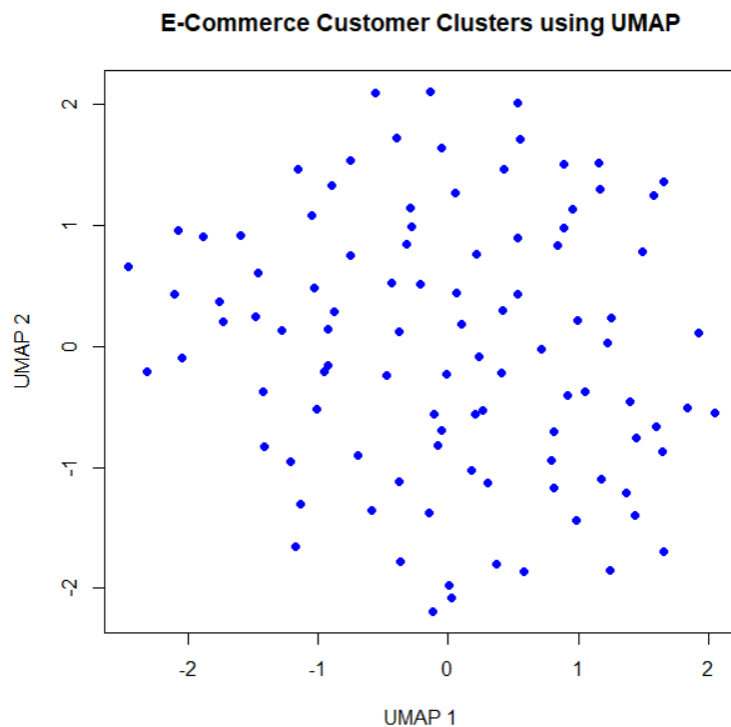
set.seed(123)

customers <- matrix(
  rnorm(100 * 60),
  nrow = 100,
  ncol = 60
)

```

```
result <- umap(customers)
plot(result$layout[,1],
      result$layout[,2],
      col = "blue",
      pch = 19,
      main = "E-Commerce Customer Clusters using UMAP",
      xlab = "UMAP 1",
      ylab = "UMAP 2")
```

OUTPUT:



Answer / Interpretation

UMAP reduces the 60 purchasing attributes into two dimensions. Customers located close to each other have similar purchasing behavior. Clearly separated groups of points indicate potential customer clusters.